

Technical University of Denmark
Written examination date: May 17, 2022.
Course Title: Computational Data Analysis
Course number: 02582
Aids allowed: Written aids allowed
Exam Duration: 4 hours
Weighting: 60%

Additional notes:

The examination is a combination of a multiple-choice test and open questions.

For the multiple choice questions, one or more answers should be given for question. Only one combination of answers is correct.

There are 20 multiple choice questions. The numbering of the individual questions is given as (1), (2), ..., (20). 2 points are given for a correct answer and 0 point is given for a wrong answer. If a question is unanswered 0 point is given.

The number of points corresponding to specific marks or needed to pass the examination is ultimately determined during censoring.

IMPORTANT NOTE: Not all suggested answers are necessarily meaningful. However, under all circumstances, among the choices you are provided for each question, there is one and only one combination as the correct answer.

There are two open questions. They are numbered (21) and (22). You can write your answers on separate papers and hand them in. Remember to mark each paper with your name, student number, the page number, and out of how many pages. Also mark each answer with the question number the answer belongs to. Answers for each open question is given from 0 up to 10 points.

Question (1)

Which method(s) are based on probabilistic models of data?

- A. ☐ Support Vector Machines
- B. ☐ K-means clustering
- C. ☐ Gaussian Mixture Models
- D. ☐ Logistic Regression
- E. ☐ None of the above

Question (2)

Which method(s) can handle data with fewer observations than dimensions?

- A. ☐ Support Vector Machines
- B. ☐ Logistic regression without regularization
- C. ☐ Random forest
- D. ☐ Principal Component Analysis
- E. ☐ None of the above

Question (3)

Which of the following statements are true for the Linear Discriminant Analysis (LDA) method?

- A. ☐ LDA is a linear method because we assume the separation function between the classes are linear
- B. ☐ LDA handles outliers better than logistic regression
- C. ☐ LDA is a probabilistic method
- D. ☐ LDA is a linear method because we assume the classes in our data are Gaussian and have the same covariance matrices
- E. ☐ None of the above

Question (4)

You happen to select a too large λ in your ridge regularization,

$$\arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2$$

How will that affect the estimated model?

- A. ☐ It will have high bias.
- B. ☐ It will have high variance
- C. ☐ Not possible to say
- D. ☐ It will have low bias
- E. ☐ None of the above

Question (5)

You happen to select a too small λ in your lasso regularization,

$$\arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

How will that affect the estimated model?

- A. ☐ It will have high bias.
- B. ☐ It will have high variance
- C. ☐ Not possible to say
- D. ☐ It will have low bias
- E. ☐ None of the above

Question (6)

How are the estimates of a linear model with lasso regularization calculated?

- A. ☐ $(X^T X + \lambda I)^{-1} X^T Y$
- B. ☐ $(X^T X)^{-1} X^T Y$
- C. ☐ Solved numerically, no analytical solution
- D. ☐ $\arg \min_{\beta} \|y - X\beta\| + \lambda \|\beta\|_2^2$
- E. ☐ None of the above

Question (7)

Is the Support Vector Machine a linear or a non-linear classifier?

- A. ☐ Linear
- B. ☐ Non-linear
- C. ☐ That depends on the chosen kernel
- D. ☐ SVM is not a classifier
- E. ☐ None of the above

Question (8)

Which techniques are developed to handle the multiple testing problem?

- A. ☐ Akaike Information Criteria
- B. ☐ Bonferroni correction
- C. ☐ Bootstrapping
- D. ☐ Benjamini-Hochberg's algorithm
- E. ☐ None of the above

Question (9)

When dealing with regularization we usually do not penalize the intercept. Why not (which statements are correct)?

- A. ☐ Penalizing the intercept would introduce bias without any reduction in variance.
- B. ☐ Penalizing the intercept would introduce variance without any reduction in bias.
- C. ☐ Penalizing the intercept would introduce both bias and variance.
- D. ☐ The model will get a lower Expected Prediction Error if we do not penalize the intercept.
- E. ☐ None of the above

Question (10)

Which of the following statements are correct?

- A. ☐ Bonferroni correction reduces the risk of false positives.
- B. ☐ Bonferroni correction reduces the chance of accepting a null hypothesis.
- C. ☐ Corrections based on a 5% False Discovery Rate is expected to give more significant findings compared to Bonferroni correction with a 5% significance level.
- D. ☐ Corrections based on a 5% False Discovery Rate is expected to give more false positives compared to Bonferroni Correction with a 5% significance level.
- E. ☐ None of the above

Question (11)

What is $\|\beta\|_2^2$, with $\beta = (\beta_1, \dots, \beta_n)^T$, equal to?

- A. ☐ $\sum_i |\beta_i|$
- B. ☐ $\sum_i \beta_i^2$
- C. ☐ $\beta^T \beta$
- D. ☐ $\max_i \beta_i^2$
- E. ☐ None of the above

Question (12)

One could argue that Self Organizing Maps are especially good at illustrating one of the blessings of dimensionality (Donoho 2000). Which one?

- A. ☐ Features will be correlated.
- B. ☐ Informative data will lie on a low dimensional manifold.
- C. ☐ Underlying structure in data will give approximative finite dimensionality.
- D. ☐ Curses and blessings of dimensionality is only an issue for supervised learning.
- E. ☐ None of the above

Question (13)

You are evaluating your new Covid-19 test and you have obtained the following confusion matrix,

		Predicted	
		Covid	No Covid
Actual	Covid	99%	1%
	No Covid	2%	98%

Now we are testing in a population of 10 000 subjects where we expect 100 to have Covid-

How many subjects do we expect to have a positive test?

- A. ☐ 99
- B. ☐ 100
- C. ☐ 200
- D. ☐ 297
- E. ☐ 300

Question (14)

We simulate data with 50 variables and 100 observations, where each variable is simulated from a random standard normal Gaussian, and the response variable is given as a weighted sum of all 50 variables, with weights $1/\sqrt{50}$, and white noise added.

For a Random Forest ensemble model with the hyper parameter of random selection of variables, $m = 5$ variables, what can you expect to hold for the bias and variance?

- A. ☐ The variance component is lower than the bias component in the expected prediction error of the ensemble.
- B. ☐ The variance component is higher than the bias component in the expected prediction error of the ensemble.
- C. ☐ The variance and bias components in the expected prediction error of the RF are of the same size.
- D. ☐ The variance of a single tree is larger than the variance of the ensemble.
- E. ☐ None of the above.

Question (15)

Figure 1 illustrates 6 two-dimensional simulated datasets and their Archetypical Analysis solution with two archetypes.

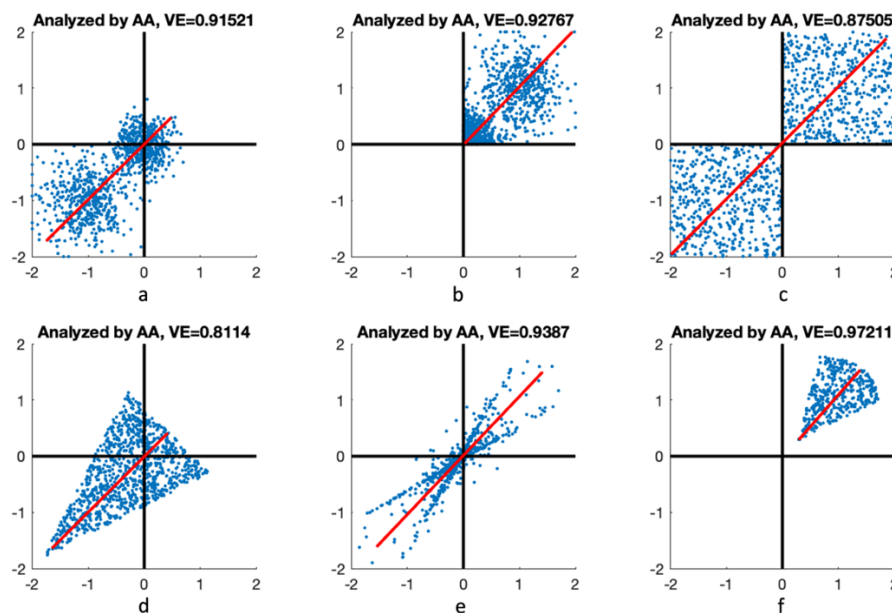


Figure 1

Which of the following statements are true:

- A. ☐ Archetypical Analysis with two components describes more of the variance in dataset c, than in any of the other datasets.
- B. ☐ Singular Value Decomposition (SVD) with two components describes all the variance for all the datasets.
- C. ☐ K-means clustering with three components is an appropriate choice for datasets d and f.
- D. ☐ K-means clustering with two components is an appropriate choice for datasets a, b, and c.
- E. ☐ None of the above

Question (16)

Which of the following statements about the Core Consistency Diagnostic (CORCONDIA) given by:

$$\text{CORCONDIA} = 100 \cdot \left(1 - \frac{\|\mathcal{J} - \mathcal{G}\|_{\text{F}}^2}{\|\mathcal{J}\|_{\text{F}}^2} \right)$$

are true?

- A. ☐ $\mathcal{J}$ is a diagonal matrix.
- B. ☐ $\mathcal{G}$ is the core tensor obtained from a Tucker decomposition of the multiway data matrix $\mathcal{X}$ of interest.
- C. ☐ CORCONDIA basically measures whether we should choose a PARAFAC or a Tucker model for our data.
- D. ☐ We choose the minimum value of CORCONDIA to select the number of components to use in PARAFAC.
- E. ☐ None of the above.

Question (17)

Which of the following statements are true for the subspace methods using derived input directions/latent variables:

- A. ☐ Principal component analysis, partial least squares, and canonical correlation analysis ALL results in latent variables constructed of linear combinations of the input variables.
- B. ☐ Principal component analysis, partial least squares, and canonical correlation analysis ALL imposes orthogonality between the produced latent variables.
- C. ☐ Principal component analysis, partial least squares, and canonical correlation analysis are ALL unsupervised methods.
- D. ☐ It is possible to use the elastic net to produce sparse versions of ALL the methods: Principal component analysis, partial least squares, and canonical correlation analysis.
- E. ☐ None of the above.

Question (18)

We have measured accelerometer data from a wearable for 45 individuals between 30 and 55 (our target group) over 7 weeks. We want to develop a model that can predict the activity level for the included as well as new individuals next week. For simplicity we sum up the motion data over each day and use a boosting algorithm to predict the total activity level for a new day. As input features we use activity levels for the past 7 days (for each individual) as well as age, gender, job function and distance to work. As response we use the activity level for each day with a history of 7 previous recording days. Given our aim, which of the following method(s) can help us assess the prediction accuracy of our boosting model?

- A. ☐ A 5-fold cross validation
- B. ☐ A leave-one-observation-out cross validation
- C. ☐ A leave-five-individuals-out cross validation
- D. ☐ A leave-one-week-out test set
- E. ☐ None of the above

Question (19)

Not part of curriculum in 2023, thus the questions is removed for the exam preparation purpose.

Question (20)

Which of the following methods are suitable to use as the individual models in a boosting ensemble?

- A. ☐ K-Nearest Neighbors with a high number of K
- B. ☐ K-Nearest Neighbors with a low number of K
- C. ☐ Any kind of classification or regression tree
- D. ☐ Any kind of linear model
- E. ☐ None of the above

NOTE: The following two questions are open-ended questions.

Question (21):

Describe the steps in the Random Forest Algorithm and how they contribute to the performance of the algorithm.

Question (22)

You are given a dataset consisting of images of faces that have been taken in the passport control in Copenhagen Airport. Security would like to know how many unique people have entered Denmark and compare it to the unique passport numbers that entered in the same time-period, as they fear a systematic fraud is happening. Which methodology, and why, would you use to analyze the data and find the number of unique people from the available face images to help security?